Institute of Information Technology
University of Dhaka
Bachelor of Science in Software Engineering (BSSE)
1st Year 1st Semester Final Examination, 2025
CSE 1101L : Structured Programming Lab
Marks: 30 # Duration: 1 hour 30 minutes

[Please read the entire question carefully and answer all of them.]

1. Write a C program that reads two integers representing a month number (1–12) and a year. Your program should determine and print the total number of days in the given month. If the entered month number is not valid, print "Invalid Month". **10**

   For example, if the input is 4 2025, the program should print 30 because April has 30 days. If the input is 1 2026, the program should print 31 because January has 31 days. Similarly, if the input is 2 2023, the program should print 28.

2. Write a C program that reads an integer N, followed by N integers representing an array. Your program must determine the frequency of each distinct element in the array and display a horizontal histogram. Each distinct element should be printed only once in ascending order, followed by a colon (:) and a sequence of asterisk (*) characters equal to the number of times that element appears in the array. Every distinct element should be printed on a separate line. **10**

   For example, if the input is N = 12 and the array is 5 2 5 3 2 5 1 3 3 2 4 1, the output should be
   1: **
   2: ***
   3: ***
   4: *
   5: ***

3. Write a C program that reads a positive integer from the user. Implement a recursive function that repeatedly replaces the number with the sum of its digits until a single-digit number is obtained. The recursive function should return the final single-digit value, and the main function should display the result. **10**

   For example, if the input is 9875, the sum of its digits is 9 + 8 + 7 + 5 = 29. Since 29 is not a single-digit number, the process is repeated by calculating 2 + 9 = 11, and then 1 + 1 = 2. Therefore, the program should print 2 as the final output.